CARTER GRIMMEISEN

SOFTWARE ENGINEER

CONTACT

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EDUCATION

MS COMPUTER SCIENCE
In progress - 4.0 GPA
University of Alabama Huntsville

BS COMPUTER SCIENCE
May 2022 - 4.0 GPA
University of Alabama Huntsville

SKILLS

PROGRAMMING LANGUAGES
Python, TypeScript, Bash, Rust, Java/Kotlin, C/C++, C#, Ada, HTML/CSS/JS

FRAMEWORKS AND RUNTIMES
Node.js, Express.js, Flask, Plotly Dash, MPI, React, React Native, Vue.js, Svelte, Electron, Flutter

DEVELOPMENT TOOLS
Git, GitLab, GitHub, Docker, Docker Compose, CI/CD, PostgreSQL, SQLite, MongoDB, Conda, Cython

CERTIFICATIONS

- Linux+
- LPIC-1 Linux Administrator
- Security+
- Network+

EXPERIENCE

SOFTWARE ENGINEER II — JUNE 2022 - PRESENT

CFD RESEARCH CORP.

+ Directed development of Android/iOS application visualizing blast overpressure exposure for service members, delivering custom 2D and 3D scenario visualizations **(React Native, TypeScript, VTK.js)**
+ Orchestrated licensing, compilation, and packaging workflows to streamline deployment of internal Python applications and libraries **(Python, Docker)**
+ Led cross-functional team to commercialize academic space-weather simulation toolkit, transitioning research into production-ready software **(Docker, Python)**
+ Built and deployed web-based dashboard to manage and execute alloy modeling simulations, improving accessibility and usability **(Python, TypeScript, SQL)**
+ Engineered real-time, web-based event dashboard for monitoring and configuring a software-defined firewall **(SvelteKit, JavaScript, SQL)**
+ Optimized model training workflows by implementing custom SQLite-backed PyTorch dataloader, accelerating research throughput **(Python, SQL)**

SOFTWARE ENGINEER - INTERN — MAY 2020 - MAY 2022

NORTHROP GRUMMAN

+ Created Ada mocking framework to enable unit testing within legacy codebase, improving test coverage and maintainability **(Ada)**
+ Automated and documented end-to-end metrics reporting workflow, enhancing reliability and reducing manual effort **(Bash, Python)**
+ Earned employee recognition award for developing automation that streamlined complex email review process, saving hundreds of engineering hours weekly **(PowerShell)**

SOFTWARE DEVELOPER — MARCH 2017 - MAY 2020

AVILUTION, LLC.

+ Implemented cryptographic signing for flight-critical configurations, enhancing security and integrity of avionics systems **(C, ECDSA)**
+ Contributed to core avionics software development, improving reliability and performance of flight systems **(C)**
+ Built web-based application for aircraft management, enabling efficient data handling and operational oversight **(MongoDB, TypeScript, Vue.js)**
+ Designed and delivered desktop application for managing display layouts and graphics, streamlining user workflows **(TypeScript, Node.js, Electron, Vue.js)**

PROJECTS

ZOD PRISMA — https://www.npmjs.com/package/zod-prisma

MADE WITH TYPESCRIPT, PRISMA, ZOD

Generates Zod schemas for database models defined in your Prisma schema in order to validate user input. Over **1.5 million downloads** on NPM and nearly **900 stars** on GitHub.